

Gamma Operators for Pochhammer Kernels — Lean Blueprint

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Overview

This blueprint tracks the formalization, in Lean 4 + mathlib, of the paper *Gamma Operators for Pochhammer Kernels*. The Lean development lives in the GammaPochhammer repository; this document mirrors its mathematical content statement by statement.

The build state, as of the current commit, is:

- Theorem 1 (determinant polynomial preserves negative-rooted) is proved modulo the two Pólya–Schur axioms (Lemmas 1 and 2).
- Theorem 2 (ladder formula) is proved unconditionally.
- Theorem 3 (centered-kernel classification) is proved modulo the Schur–Maló axiom (Lemma 2).
- Lemma 3 (the γ -representation of palindromic negative-rooted polynomials) is proved from first principles.

Every *green* node in the dependency graph is fully formalized in Lean. Two red nodes remain: the two Pólya–frequency / Schur classical results that mathlib does not yet have, namely Lemmas 1 and 2.

Chapter 1

Preliminaries

1.1 Polynomials with real, non-positive zeros

Definition 1 (Real non-positive zeros). A real polynomial $p \in \mathbb{R}[X]$ is said to have *only real, non-positive zeros* when every complex root of p , regarded as a polynomial in $\mathbb{C}[X]$, lies on the non-positive real axis $\mathbb{R}_{\leq 0}$.

Definition 2 (Real strictly negative zeros). A real polynomial $p \in \mathbb{R}[X]$ has *only real, strictly negative zeros* when every complex root lies in $(-\infty, 0)$. Equivalently: $p(0) \neq 0$ and p has only real, non-positive zeros.

Definition 3 (Degree marker). We write $\text{HasDegree}(p, n)$ for the proposition that $p \in \mathbb{R}[X]$ has natural degree exactly n and is non-zero.

1.2 Pochhammer symbols and the determinant polynomial

Definition 4 (Pochhammer symbol). For $a \in \mathbb{R}$ and $k \in \mathbb{N}$, the rising factorial is

$$(a)_k = a(a+1)(a+2) \cdots (a+k-1) = \prod_{j=0}^{k-1} (a+j) \in \mathbb{R}[X].$$

Definition 5 (Linear Pochhammer). $\text{linearPochhammer}(b, a, k)$ is the polynomial obtained by substituting $bX + a$ into $(X)_k$; used to model the argument of a rung kernel.

Definition 6 (Ordinary generating polynomial). For $n \in \mathbb{N}$ and $f : \mathbb{N} \rightarrow \mathbb{R}$ we set

$$\text{ordinaryGen}(n, f) = \sum_{k=0}^n f(k) X^k.$$

Definition 7 (Hadamard product). The (truncated) Hadamard product of $A, B \in \mathbb{R}[X]$ at order n is

$$\text{hadamard}(n, A, B) = \sum_{k=0}^n a_k b_k X^k, \quad a_k = [X^k]A, \quad b_k = [X^k]B.$$

Definition 8 (Determinant kernel). $K_{n,k}^{\det}(X) = (X)_k (X)_{n-k} - (X+1)_k (X-1)_{n-k}$.

Definition 9 (Determinant polynomial P_n). Given $f : \mathbb{N} \rightarrow \mathbb{R}$ and $n \in \mathbb{N}$,

$$P_n(X; f) = \sum_{k=0}^n \binom{n}{k} f_k f_{n-k} K_{n,k}^{\det}(X).$$

1.3 Palindromic polynomials and the γ -basis

Definition 10 (Palindromic of degree n). $Q \in \mathbb{R}[X]$ is *palindromic of formal degree n* when its natural degree is at most n and $[X^{n-k}]Q = [X^k]Q$ for all $0 \leq k \leq n$.

Definition 11 (γ -basis). For $n, j \in \mathbb{N}$ we set $B_{n,j} = X^j(1+X)^{n-2j}$. These are the building blocks of the γ -representation of any palindromic polynomial of degree n with only real, non-positive zeros.

Definition 12 (γ -expansion). $\text{IsGammaExpansion}(n, m, Q, \gamma)$ asserts that $Q = \sum_{j=0}^m \gamma_j B_{n,j}$.

Definition 13 (γ -polynomial). $G_m(\gamma) = \sum_{j=0}^m \gamma_j X^j$, the ordinary generating polynomial of the γ -coefficients.

Chapter 2

Theorem 1: the determinant polynomial

2.1 The two Pólya–Schur axioms

These are the only two statements in the development that are still axioms. Both are classical Pólya–frequency results that mathlib does not currently provide.

Lemma 14 (Hadamard closure / Aissen–Schoenberg–Whitney). *Let $A, B \in \mathbb{R}[X]$ have non-negative coefficients up to degree n and only real, non-positive zeros. Then the truncated Hadamard product $\text{hadamard}(n, A, B)$ has only real, non-positive zeros.*

The Lean statement is recorded as the axiom `hsw` in `Basic.lean`; an axiom-free proof would require the Aissen–Schoenberg–Whitney theory of totally-positive sequences.

Lemma 15 (Schur–Maló: $\mathcal{S}$ preserves non-positive-rooted). *Let $B \in \mathbb{R}[X]$ have only real, non-positive zeros. Then for all $\ell \in \mathbb{N}$ and $\alpha > 0$, the Schur transform $\mathcal{S}_{\ell, \alpha}(B)$ has only real, non-positive zeros.*

Recorded in Lean as the axiom `hsk`. An axiom-free proof would require Hermite–Kakeya–Obreschkoff (HKO) interlacing.

2.2 Lemma 3: the γ -representation

Lemma 16 (γ -representation). *Let Q be palindromic of formal degree $n = 2m + \varepsilon$ ($\varepsilon \in \{0, 1\}$) with only real, strictly negative zeros. Then there exist coefficients $\gamma_0, \dots, \gamma_m$ such that $Q = \sum_{j=0}^m \gamma_j B_{n,j}$. Moreover the γ -polynomial $\sum_{j=0}^m \gamma_j X^j$ has only real, non-positive zeros.*

Proof. Strong induction on the degree n , using the paper’s reciprocal-pair factorization. The Lean proof is in `GammaPochhammer/GammaRep.lean` and uses no project-specific axioms. $\square$

2.3 Theorem 1

Theorem 17 (Theorem 1 — P_n preserves real-negative-rooted families). *Let $n \geq 2$ and $f : \mathbb{N} \rightarrow \mathbb{R}$ be such that the generating polynomial $\text{ordinaryGen}(n, f)$ has only real, strictly negative zeros and degree exactly n . Then $P_n(X; f)$ has only real, strictly negative zeros.*

Proof. Follow the paper’s proof of Theorem 1:

1. Form the convolution polynomial $Q = \sum_k \binom{n}{k} f_k f_{n-k} X^k$ (Lemma 19).
2. Show Q is palindromic of degree n with only real, strictly negative zeros (Lemma 20), using Hadamard closure (Lemma 14) twice and the $(1 + X)^n$ baseline.
3. Identify P_n with the $s = 1, \mu = 0$ rung operator applied to Q (Lemma 21).
4. Apply the rung-action formula (Lemma 29) together with the γ -representation (Lemma 16) to express P_n as a Schur transform of Q .
5. Conclude by Lemma 15, using $f_1 f_{n-1} > f_0 f_n$ (Lemma 23) to exclude a zero at the origin.

Full Lean proof in `GammaPochhammer/Determinant.lean`. □

2.3.1 Vieta + convolution infrastructure

These are paper-internal results used by the proof of Theorem 1. All are formalized in Lean and depend on no project-specific axiom.

Lemma 18 (Convolution polynomial). $\text{convolutionPoly}(n, f) = \sum_{k=0}^n \binom{n}{k} f_k f_{n-k} X^k$.

Lemma 19 (Convolution polynomial is palindromic). $\text{convolutionPoly}(n, f)$ is palindromic of degree n .

Lemma 20 (Convolution polynomial has only real negative zeros). *Under the hypotheses of Theorem 17, $\text{convolutionPoly}(n, f)$ has only real, strictly negative zeros. Proved by two applications of Hadamard closure (with ordinaryGen , its coefficient-reversal, and the baseline $(1 + X)^n$).*

Lemma 21 (P_n as a rung operator). $P_n(X; f) = T_{1,0}^{(n)}(\text{convolutionPoly}(n, f))$.

Lemma 22 ($P_n(0)$). $P_n(0; f) = n! (f_1 f_{n-1} - f_0 f_n)$.

Lemma 23 ($f_1 f_{n-1} > f_0 f_n$). *Under the hypotheses of Theorem 17 we have $f_1 f_{n-1} > f_0 f_n$. This is the paper’s “ $e_1 e_{n-1} > e_n$ ” Newton-type inequality and prevents P_n from acquiring a root at the origin.*

Chapter 3

Theorem 2: the Pochhammer kernel ladder

3.1 Rung kernel and rung operator

Definition 24 ($\lambda_{s,q}$ coefficients). $\lambda_{s,q} \in \mathbb{R}$ is the alternating central-difference coefficient

$$\lambda_{s,0} = \frac{1}{(s!)^2}, \quad \lambda_{s,q} = \frac{2(-1)^q}{(s-q)!(s+q)!} \quad (q > 0).$$

Definition 25 (Rung kernel $K_{n,k}^{[s,\mu]}$). $K_{n,k}^{[s,\mu]}(X) = \sum_{q=0}^s \lambda_{s,q} (X + \mu + q)_k (X + \mu - q)_{n-k}$.

Definition 26 (Rung operator $T_{s,\mu}^{(n)}$). For $Q \in \mathbb{R}[X]$, $T_{s,\mu}^{(n)}(Q) = \sum_{k=0}^n [X^k]Q \cdot K_{n,k}^{[s,\mu]}$.

3.2 The A380113 triangle

Definition 27 ($A_{s,q}$ triangle). $A_{s,q}$ is the unsigned normalized triangle from OEIS A380113: $A_{0,0} = 1$, $A_{s,0} = \frac{1}{2} \binom{2s}{s}$ for $s \geq 1$, and $A_{s,q} = \binom{2s}{s-q}$ for $1 \leq q \leq s$.

3.3 Lemma 4 and Lemma 5

Lemma 28 (Lemma 4 — central-difference identity). *The alternating sum $\sum_q \lambda_{s,q}(w+q)$ collapses to the paper's closed-form central-difference identity.*

Lemma 29 (Lemma 5 — rung action on the γ -basis). *The rung operator $T_{s,\mu}^{(n)}$ acts on a single γ -basis element $B_{n,j}$ by the explicit formula recorded in the paper (eq. $TB_{n,j}$).*

3.4 Theorem 2 and its corollaries

Theorem 30 (Theorem 2 — ladder formula). *Statement of the paper's ladder formula: applied to a γ -expansion, the rung operator $T_{s,\mu}^{(n)}$ factors as a Pochhammer prefactor*

times the Schur transform of $4^{m-s}\mathbf{G}^{(s)}(t/4)$, evaluated after the shift $X \mapsto X + \mu$. (Full statement in the source.)

Theorem 31 (Kernel ladder, downstream). *Downstream form of Theorem 2 used by Corollary 1 and Corollary 2.*

Theorem 32 (Strict-shift form). *Variant in which the shift parameter is strict.*

Corollary 33 (Corollary 1 — original determinant kernel). *On palindromic negative-rooted inputs, the original determinant-kernel operator produces a polynomial that is either zero or has only real, non-positive zeros.*

Corollary 34 (Corollary 2 — single-product kernel at $s = 0$). *At $s = 0$ the rung kernel collapses to a single Pochhammer product.*

Chapter 4

Theorem 3: classification of universally-admissible centered kernels

4.1 The centered two-product family

Definition 35 (Centered kernel $K_{n,k}^{a,c}$). $K_{n,k}^{a,c}(X) = (X+a)_k (X-a)_{n-k} - (X+c)_k (X-c)_{n-k}$.

Definition 36 (Centered operator). $T_n^{a,c}(Q) = \sum_{k=0}^n [X^k]Q \cdot K_{n,k}^{a,c}$.

Definition 37 (Universally admissible). A pair $(a, c) \in \mathbb{R}^2$ is *universally admissible* when $T_n^{a,c}$ sends every palindromic polynomial Q with only real, strictly negative zeros to a polynomial that is either zero or has only real, non-positive zeros.

4.2 Theorem 3

Theorem 38 (Theorem 3 — centered classification). *Let $a, c \in \mathbb{R}$ with $a^2 \neq c^2$. Then (a, c) is universally admissible if and only if $\{a^2, c^2\} = \{0, 1\}$ as subsets of $\mathbb{R}$.*

Proof. Follows the paper’s discriminant + intermediate-value-theorem argument: $\sigma = u + v = 1$ is forced via two λ -tests on a perturbation $Q_n^{(4)}(\lambda) = (\text{quadFactor } \lambda)^2$; $u = 0$ is then forced by an ε -test on $Q_n^{(6)}(\varepsilon) = (\text{quadFactor } \varepsilon)^3$ and the intermediate value theorem. Sufficiency reduces to the determinant-kernel case using sign and palindromic symmetries via Corollary 33. Full Lean proof in `GammaPochhammer/Classification.lean`. $\square$

Bibliography